

TASKS

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Task 1:

Install Visual C++ and attach proper screenshots to Lab File.

Task 2:

Draw a flowchart & Write a C++ program to store statement record such as (Name, Age, Grade and CGPA).

Pseudo code:

STEP 1: START

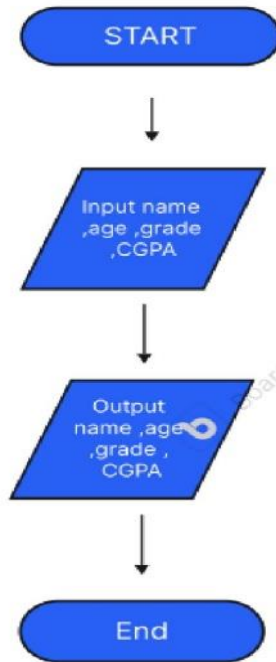
STEP 2: declare name, age, grade, CGPA

STEP 3: Input name, age, grade, CGPA

STEP 4: Output name, age, grade, CGPA

STEP 5: End

Flowchart:



C++ program:

```
#include<iostream>
#include<string>
Using namespace std;
Int main()
{
    String name,grade;
    Int age;
    Float CGPA;
    Std::cout << "enter name " << std::endl;
    Std::cin >> name;
    Std::cout << "enter age" << std::endl;
    Std::cin >> age;
    Std::cout << "enter grade" << std::endl;
    Std::cin >> grade;
    Std::cout << "enter CGPA" << std::endl;
    Std::cin >> CGPA;
    Return 0;
```

The screenshot shows a terminal window titled 'input'. The user has entered the following data: name 'Javaria', age '18', grade 'A+', and CGPA '3.9'. The program has finished with exit code 0, and the user is prompted to press ENTER to exit the console.

```
input
enter name
Javaria
enter age
18
enter grade
A+
enter CGPA
3.9
...Program finished with exit code 0
Press ENTER to exit console.
```

}

Task 3:

Draw a flowchart and Write a C++ program to read student's three grade, calculate the average of the grade, and then display the average grade.

Pseudo code:

Step 1: start

Step 2: declare g1, g2, g3, avg

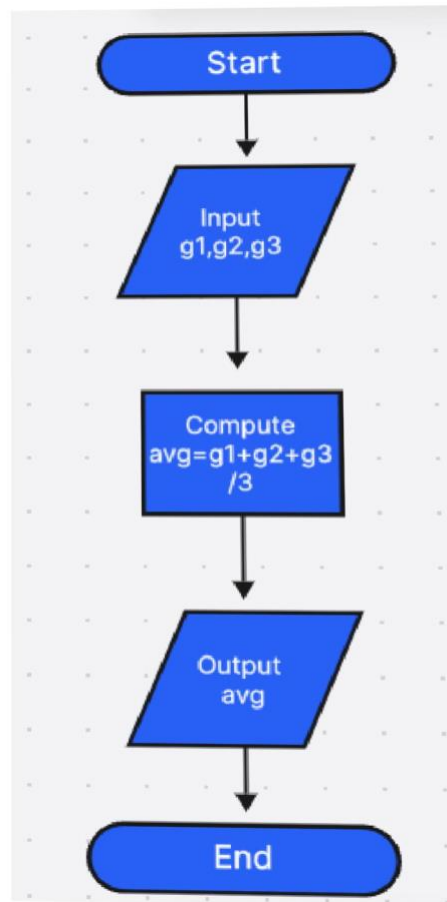
Step 3: input g1, g2, g3

Step 4: compute avg of g1, g2, g3

Step 5: output avg

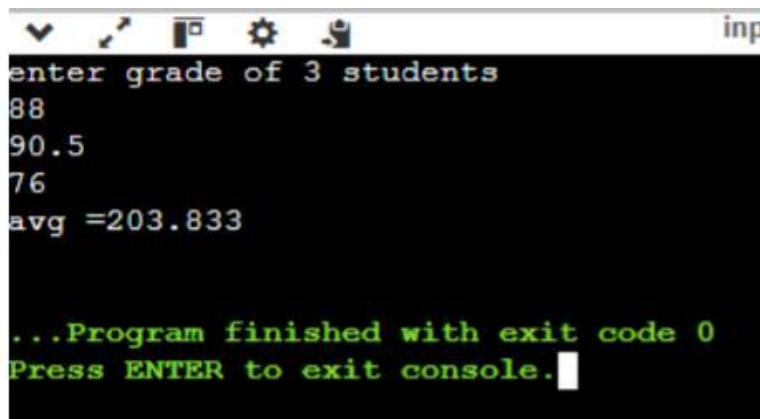
Step 6: end

Flowchart:



C++ program:

```
#include<iostream>
Using namespace std;
Int main()
{
    Float g1,g2,g3,avg;
    Std::cout << "enter grade of 3 students" << std::endl;
    Std::cin >> g1 >> g2 >> g3;
    Avg = g1+g2+g3/3;
    Std::cout << "avg =" << avg << std::endl;
    Return 0;
}
```



```
inp
enter grade of 3 students
88
90.5
76
avg =203.833

...Program finished with exit code 0
Press ENTER to exit console.
```

Task 4:

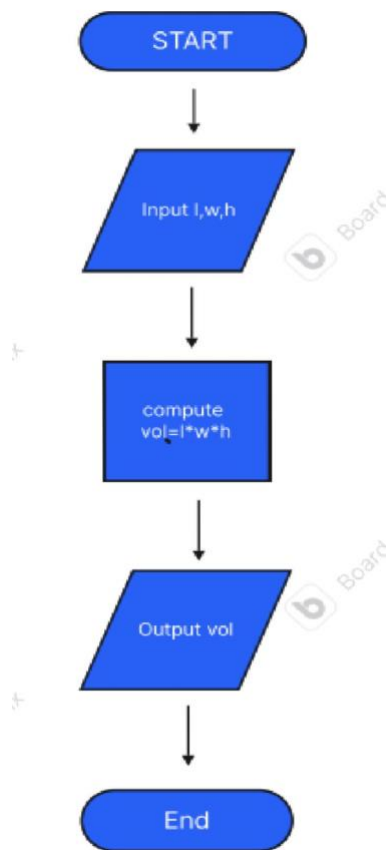
Write program & draw a flowchart that reads the height, length, and width of the rectangular box, Calculates and displays the volume.

Note: Volume = lwh.

Pseudo code:

- Step 1: start
- Step 2: declare l, w, h, vol
- Step 3: input l, w, h
- Step 4: compute $vol = l * w * h$
- Step 5: output vol
- Step 6: end

Flowchart:



C++ program:

```
#include<iostream>
Using namespace std;
Int main()
{
    Float l,w,h,vol;
    Std::cout << "enter l,w,h " << std::endl;
    Std::cin >> l >> w >> h;
    Vol=l*w*h;
    Std::cout << "vol =" << vol << std::endl;
    Return 0;
}
```

```
enter l,w,h
8
5
4.3
vol =172

...Program finished with exit code 0
Press ENTER to exit console.
```

A screenshot of a terminal window with a dark background. The prompt 'enter l,w,h' is shown. The user has entered the values 8, 5, and 4.3 on separate lines. The program outputs 'vol =172'. At the bottom, it shows '...Program finished with exit code 0' and 'Press ENTER to exit console.' with a cursor.

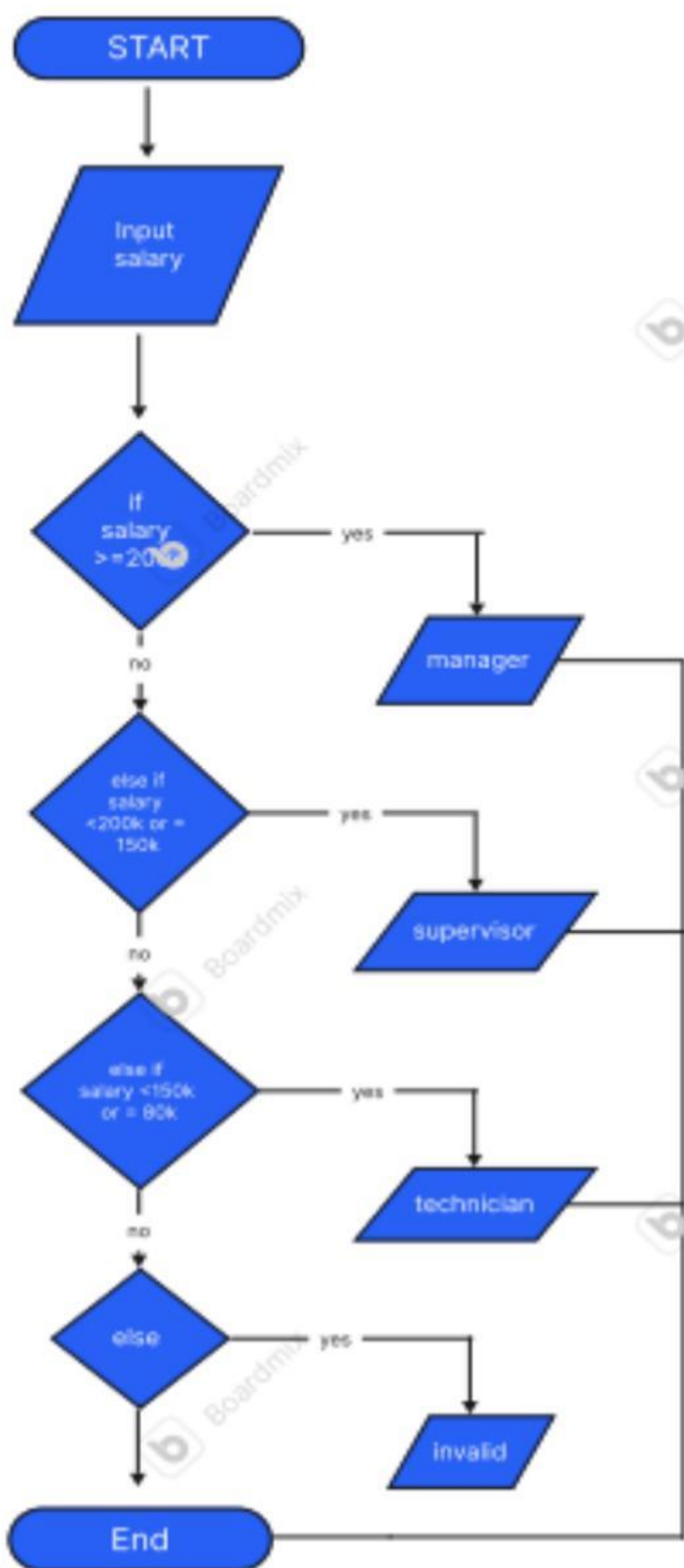
Task 5:

Write a program to read the salary of an employee and prints particular designation of the concern person for example Manager salary = 200k, Supervisor = 150k and Technician = 80k.

Pseudo code:

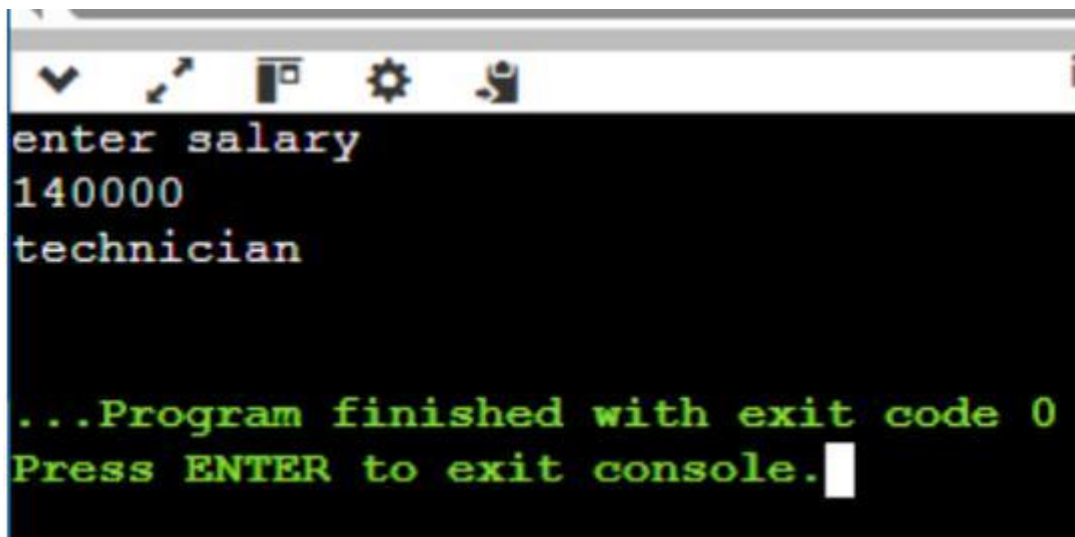
Step 1: start
Step 2: declare salary
Step 3: input salary
Step 4: if salary $\geq 200k$ then print manager
 Else if salary $\leq 200k$ or $= 150k$ then print supervisor
 Else if salary $\leq 150k$ or $= 80k$ then print technician
 Else invalid
Step 5: end

Flowchart:



C++ program:

```
#include<iostream>
Using namespace std;
Int main()
{
    Int salary;
    Std::cout << "enter salary" << std::endl;
    Std::cin >> salary;
    If(salary>=200000)
    Std::cout << "manager" << std::endl;
    Else if(salary<200000&&salary>=150000)
    Std::cout << "supervisor" << std::endl;
    Else if(salary<150000&&salary>=80000)
    Std::cout << "technician" << std::endl;
    Else
    Std::cout << "invalid salary" << std::endl;
    Return 0;
}
```

A screenshot of a console window showing the execution of a C++ program. The window has a title bar with standard icons. The console output is as follows: 'enter salary' followed by a newline, '140000' followed by a newline, and 'technician' followed by a newline. Below this, in green text, it says '...Program finished with exit code 0' and 'Press ENTER to exit console.' with a white cursor at the end of the second line.

```
enter salary
140000
technician

...Program finished with exit code 0
Press ENTER to exit console.
```